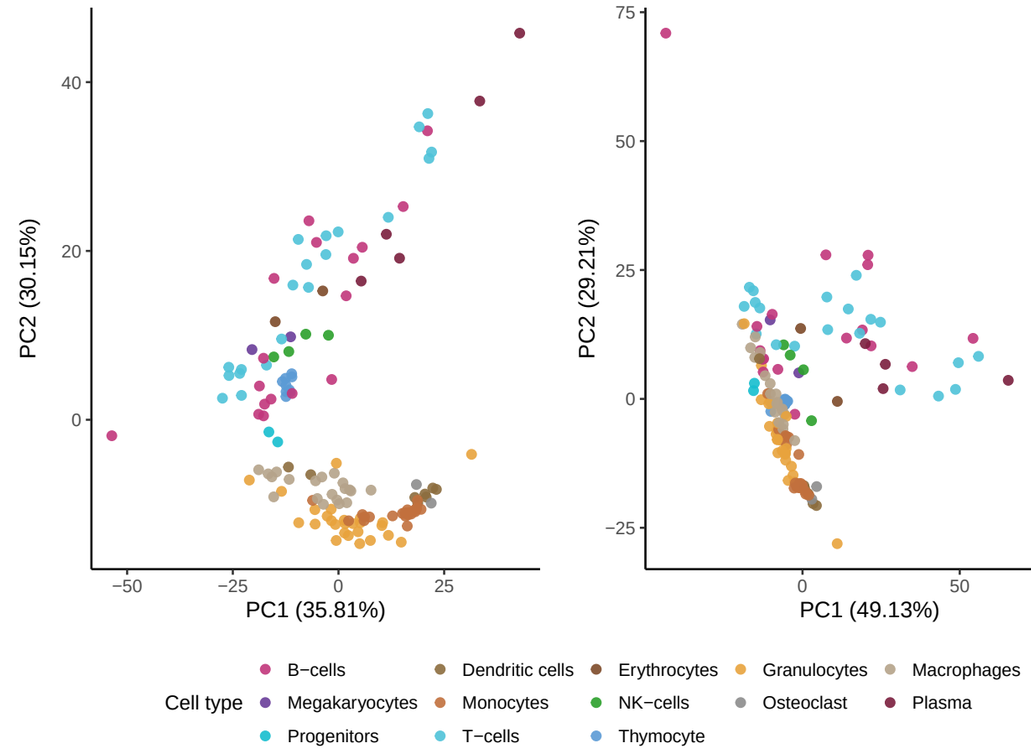
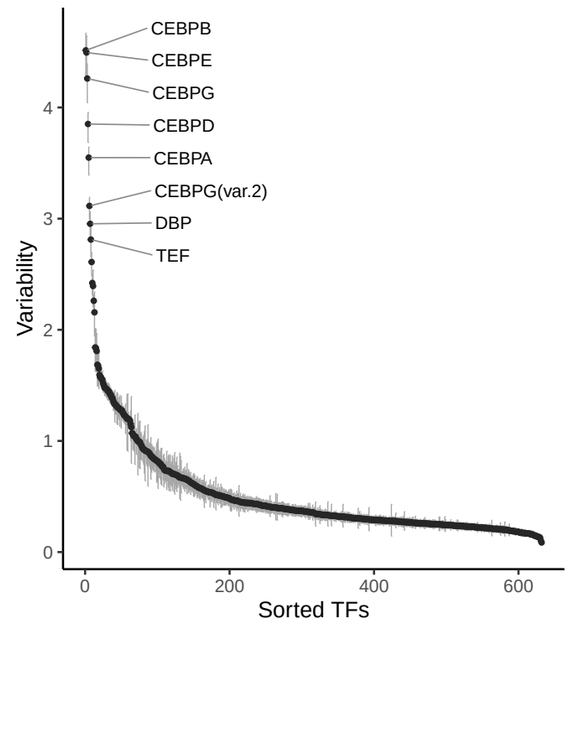
A


Uncorrected deviations (JASPAR2020 distal)
cell type R2: PC1 0.45, PC2 0.80
PC2 (30.15%)
PC1 (35.81%)

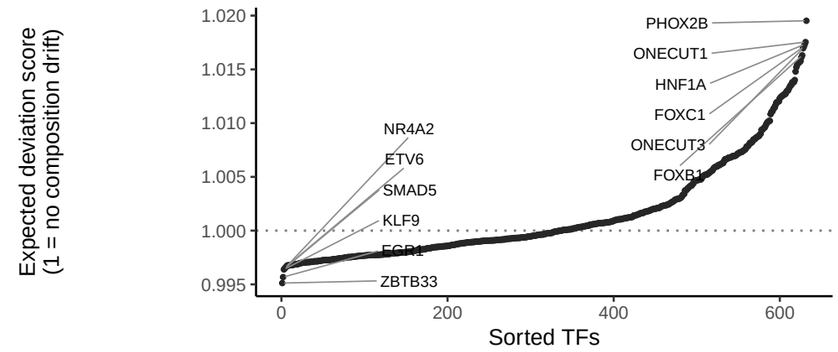
B


Bias corrected deviations (JASPAR2020)
cell type R2: PC1 0.40, PC2 0.63
PC2 (29.21%)
PC1 (49.13%)

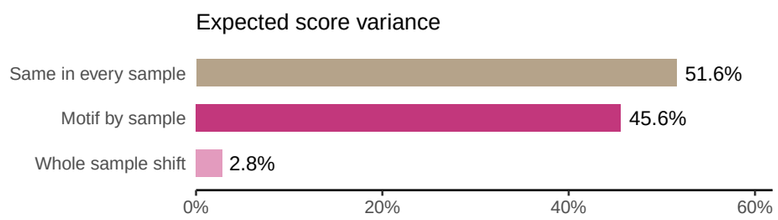
C

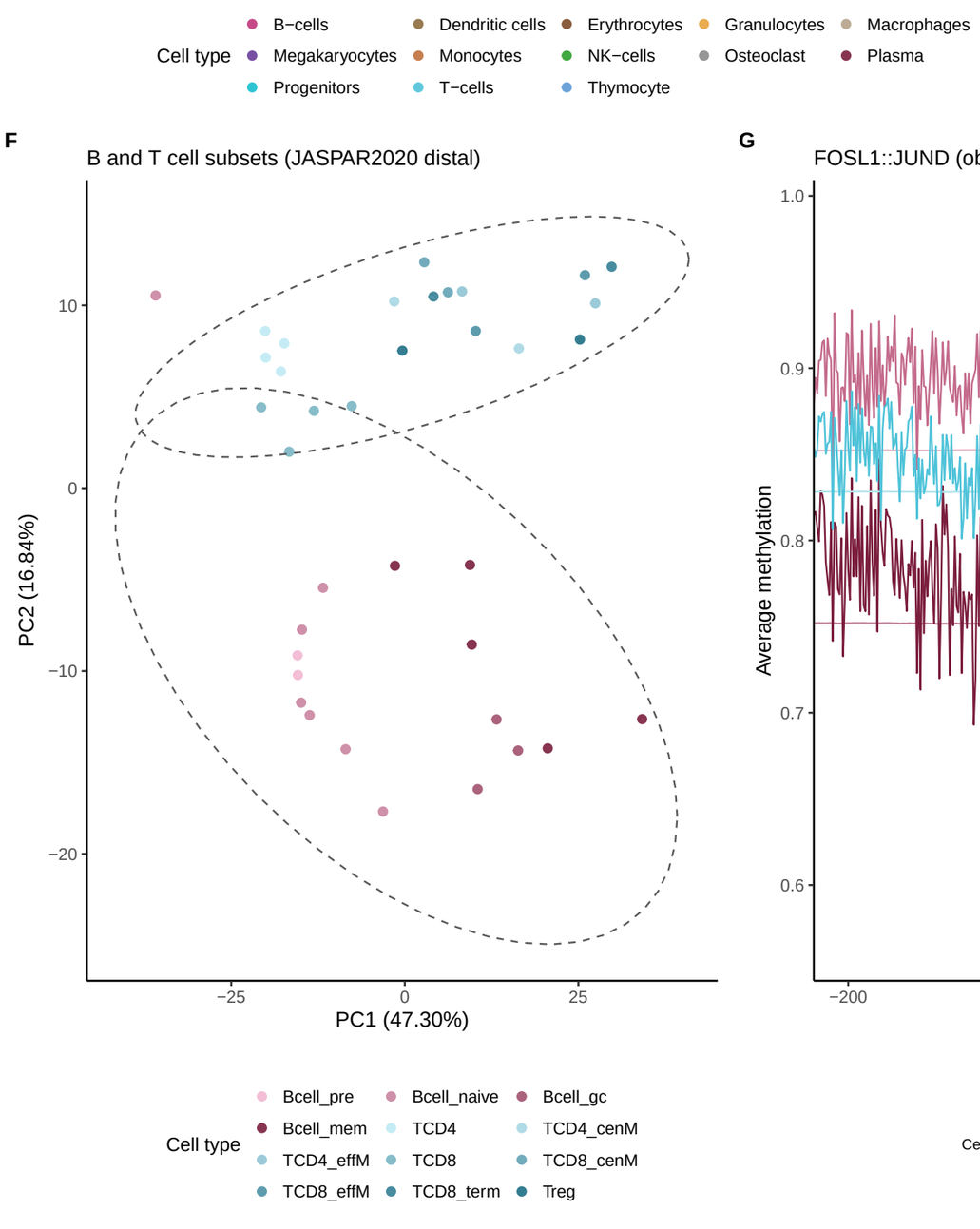

Motif variability jaspar2020_distal
CEBPB
CEBPE
CEBPG
CEBPD
CEBPA
CEBPG(var.2)
DBP
TEF
Variability
Sorted TFs

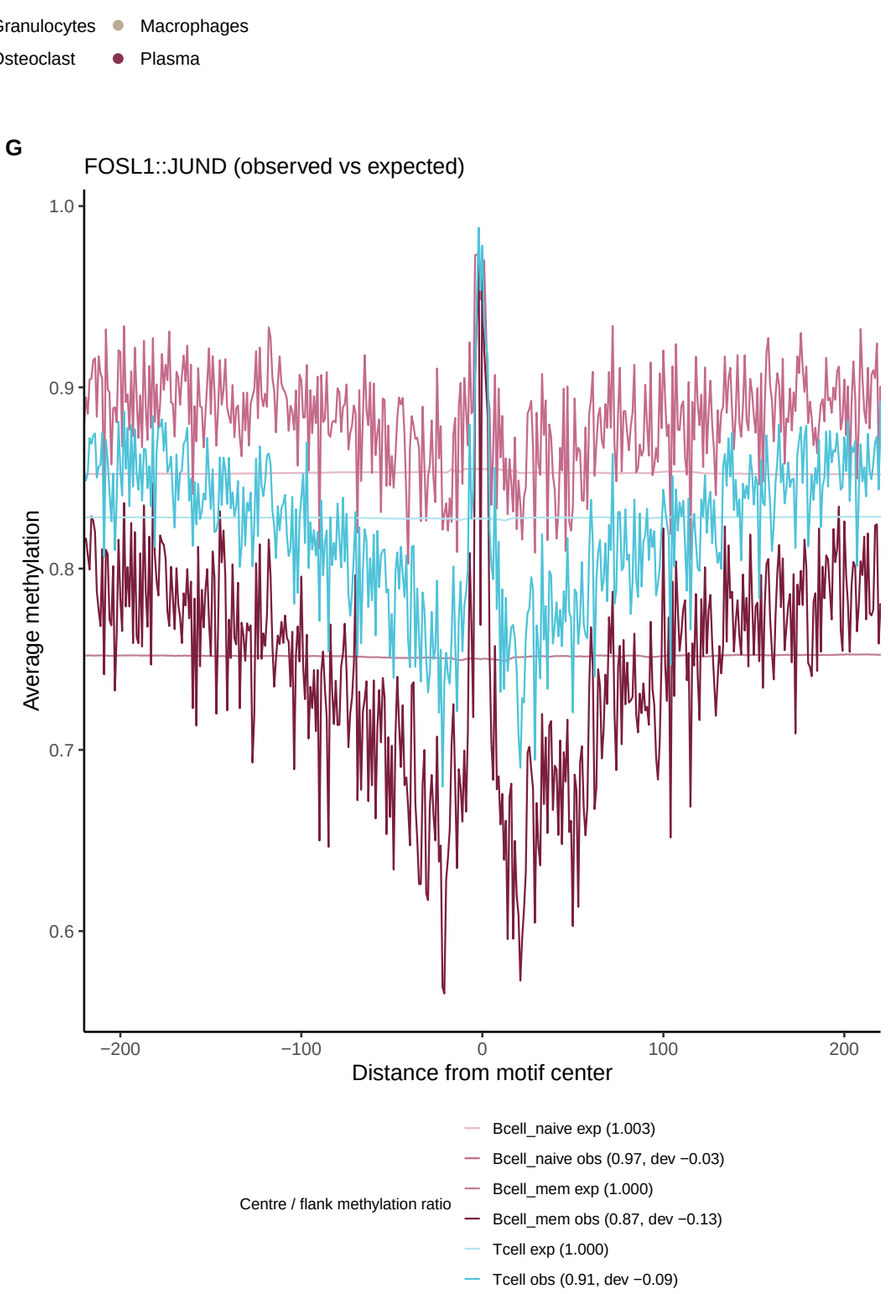
D

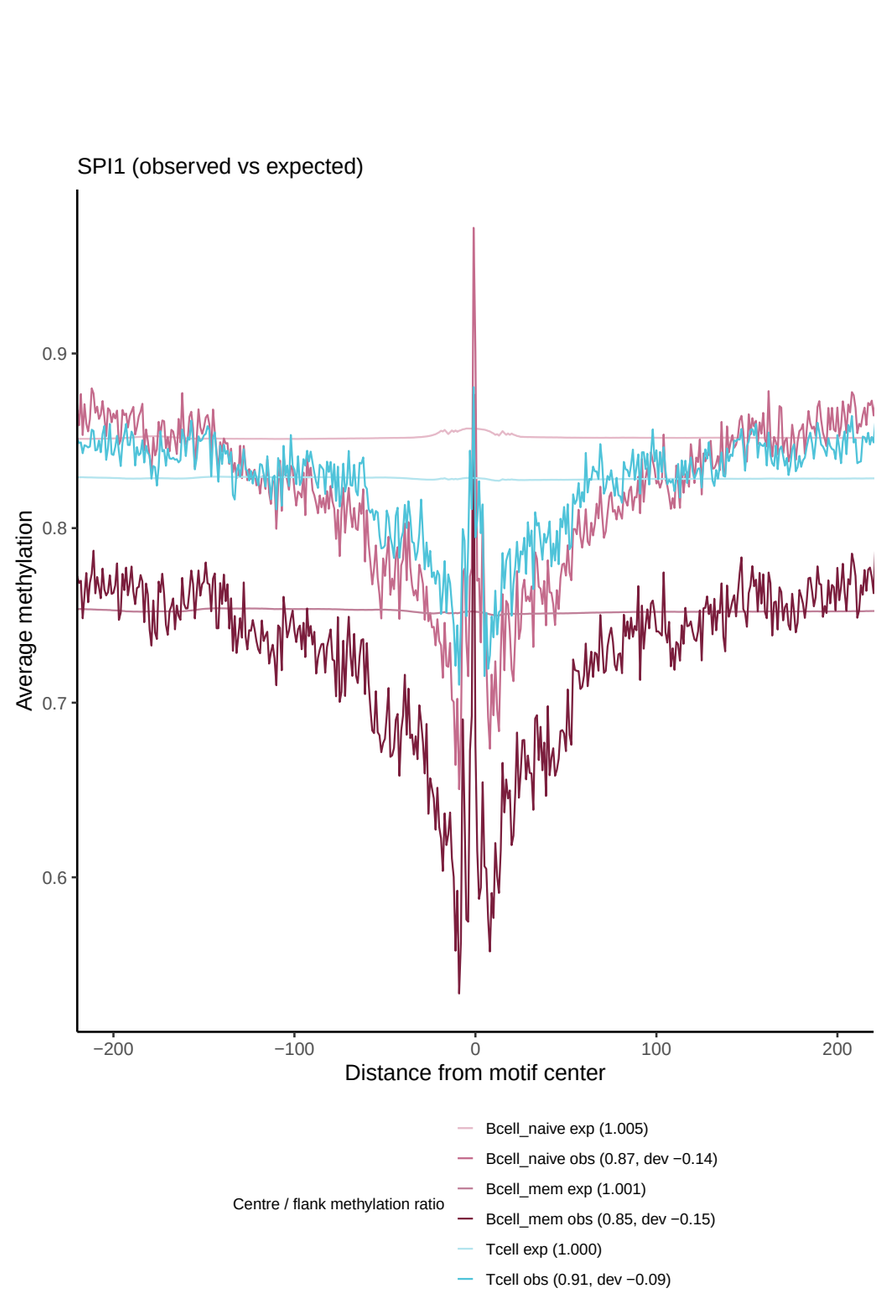

Expected deviation scores jaspar2020_distal
Expected deviation score
(1 = no composition drift)
PHOX2B
ONECUT1
HNF1A
FOXC1
ONECUT3
FOXB1
NR4A2
ETV6
SMAD5
KLF9
EGR1
ZBTB33
Sorted TFs

E

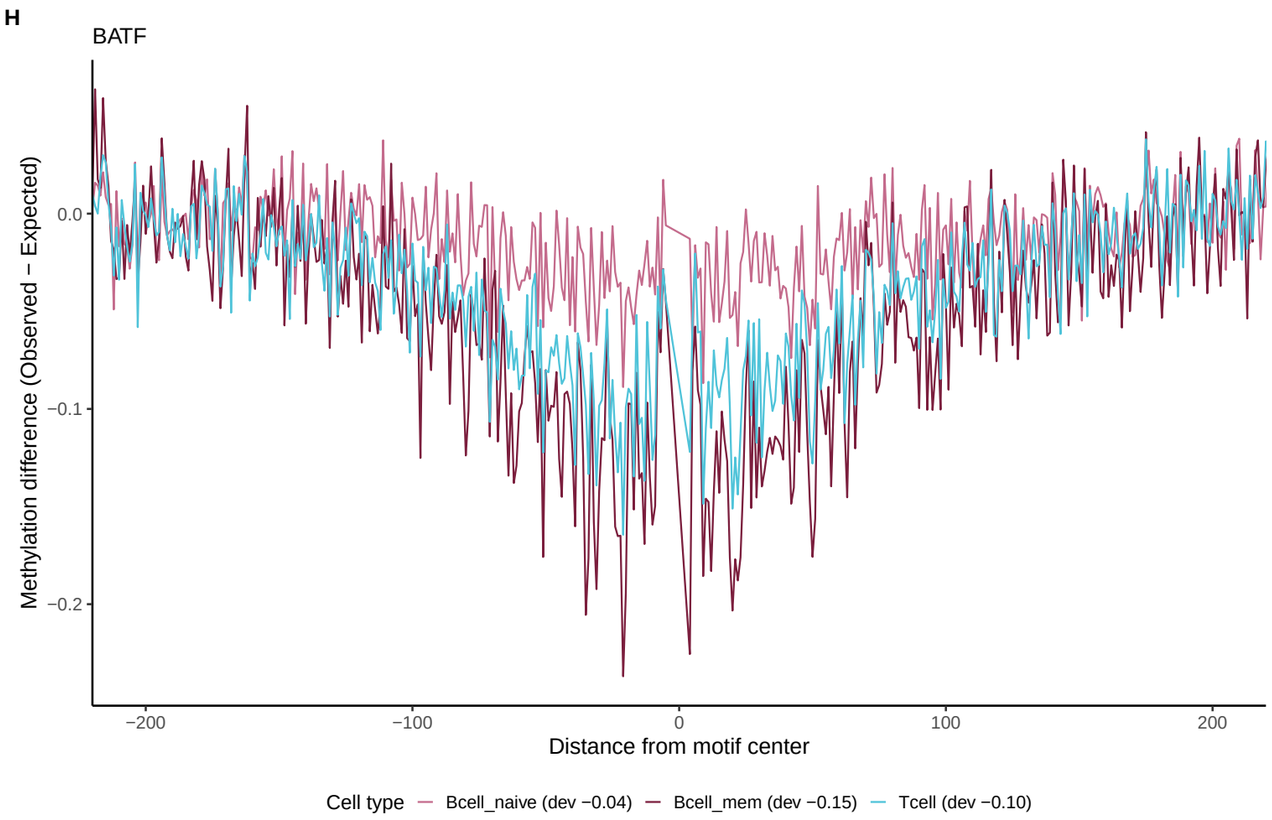

Expected score variance
Same in every sample
51.6%
Motif by sample
45.6%
Whole sample shift
2.8%

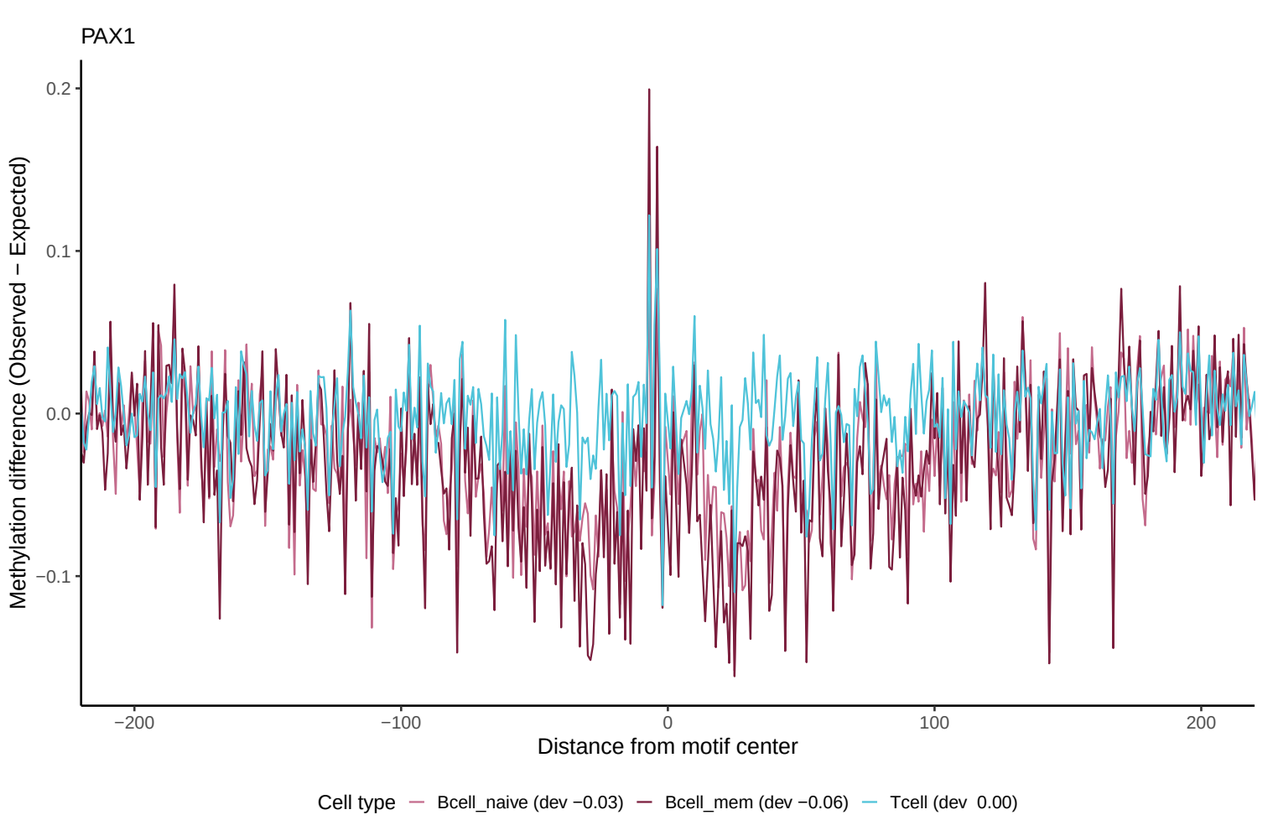

F
B and T cell subsets (JASPAR2020 distal)
PC2 (16.84%)
PC1 (47.30%)
Cell type
Bcell_pre
Bcell_naive
Bcell_gc
Bcell_mem
TCD4
TCD4_cenM
TCD4_effM
TCD8
TCD8_cenM
TCD8_effM
TCD8_term
Treg


G
FOSL1::JUND (observed vs expected)
Average methylation
Distance from motif center
Centre / flank methylation ratio
Bcell_naive exp (1.003)
Bcell_naive obs (0.97, dev −0.03)
Bcell_mem exp (1.000)
Bcell_mem obs (0.87, dev −0.13)
Tcell exp (1.000)
Tcell obs (0.91, dev −0.09)


SPI1 (observed vs expected)
Average methylation
Distance from motif center
Centre / flank methylation ratio
Bcell_naive exp (1.005)
Bcell_naive obs (0.87, dev −0.14)
Bcell_mem exp (1.001)
Bcell_mem obs (0.85, dev −0.15)
Tcell exp (1.000)
Tcell obs (0.91, dev −0.09)

H


BATF
Methylation difference (Observed − Expected)
Distance from motif center
Cell type
Bcell_naive (dev −0.04)
Bcell_mem (dev −0.15)
Tcell (dev −0.10)


PAX1
Methylation difference (Observed − Expected)
Distance from motif center
Cell type
Bcell_naive (dev −0.03)
Bcell_mem (dev −0.06)
Tcell (dev 0.00)